

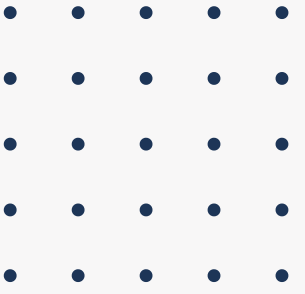
Data 144: Data Mining Final

Customer Segmentation and Sales Forecasting for Grocery Retail

by:
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Introduction



Motivation:

Grocery retail generates large, high-dimensional transactional data with strong behavioral patterns.

Customer behavior is heterogeneous, making global models perform poorly without segmentation.

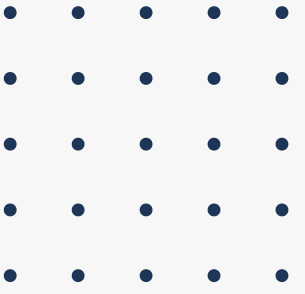
This motivates a **cluster-then-predict framework** combining unsupervised and supervised learning for grocery retail.



Stakeholders:

Our stakeholders include **retail managers, category planners, marketing teams, and executives** who rely on accurate demand predictions and customer insights.

By providing segment-level understanding and tailored forecasting, they can **plan inventory better, target promotions more effectively, and retain high-value customers.**



Research Questions:

1

Can we discover meaningful **customer segments** using shopping behavior?

2

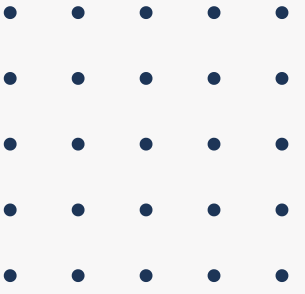
Can **segment-specific ML models** improve:

- next-category predictions,
- and churn identification?

3

How can these insights help retailers make **better decisions**?

Introduction



Source & Structure of our Dataset:

- Kaggle: **Grocery Sales Dataset**

- [Kaggle Link](#)

- Relational database with **6 tables**:

- customers.csv
 - sales.csv
 - products.csv
 - categories.csv
 - cities.csv
 - countries.csv

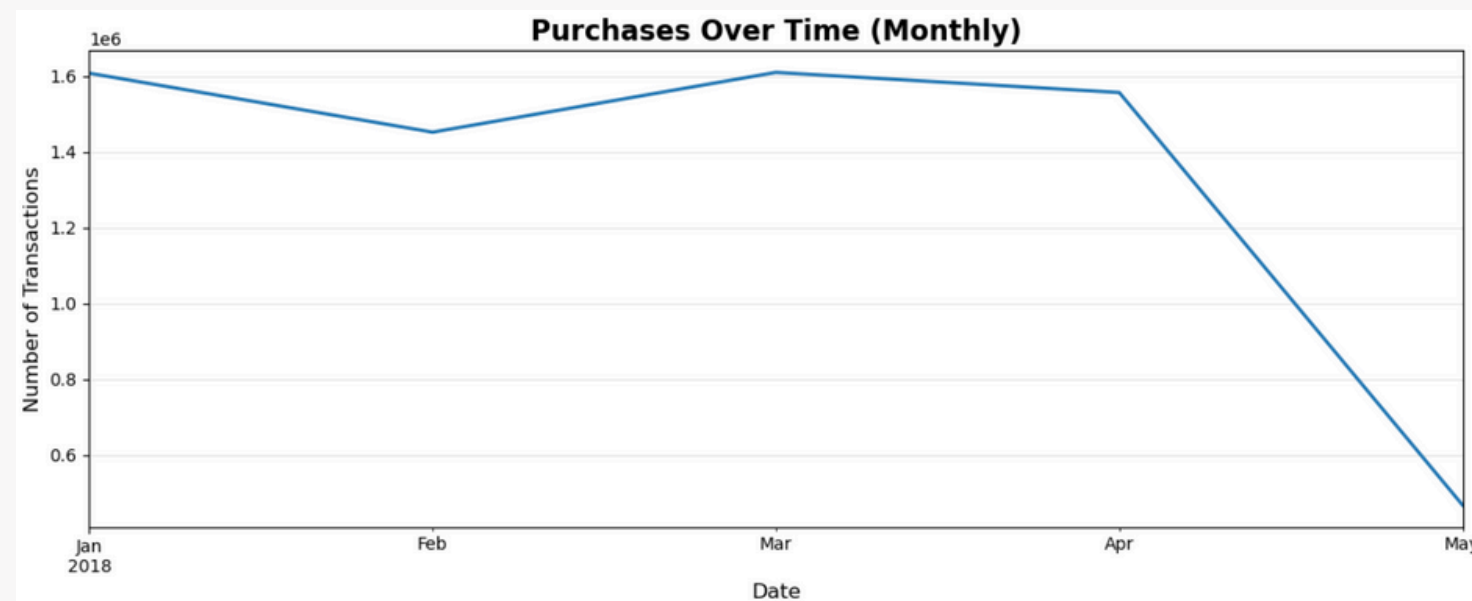
	SalesID	SalesPersonID	CustomerID	ProductID	Quantity	Discount	TotalPrice	SalesDate	TransactionNumber	Price	CategoryID	CategoryName
0	1	6	27039	381	7	0.0	309.6359	2018-02-05 07:38:25.430	FQL4S94E4ME1EZFTG42G	44.2337	1	Confections
1	2	16	25011	61	7	0.0	437.8220	2018-02-02 16:03:31.150	12UGLX40DJ1A5DTFBHB8	62.5460	8	Grain
2	3	13	94024	23	24	0.0	1896.4416	2018-05-03 19:31:56.880	5DT8RCPL87KI5EORO7B0	79.0184	11	Produce
3	4	8	73966	176	19	0.2	1545.0173	2018-04-07 14:43:55.420	R3DR9MLD5NR76VO17ULE	81.3167	6	Seafood
4	5	10	32653	310	9	0.0	719.8020	2018-02-12 15:37:03.940	4BGS0Z5OMAZ8NDAFHHP3	79.9780	9	Poultry

- ~2.4M sales transactions, ~98k customers, ~3k products.
(sales has 2,437,905 rows; customers has 98,759 rows)

Our Dataset

Key Descriptives:

- 2.4M sales transactions
- 98k unique customers
- 3k products across grocery categories
- Transaction fields: CustomerID, ProductID, Quantity, Discount, Price, TotalPrice, SalesDate, Category
- Time span: Start of Jan– start of May 2018
- Strong coverage for behavioral modeling



Preprocessing:

Standardization

- Converted ID columns to strings
- Parsed SalesDate
- Added TotalPrice = Quantity × Price

Filtering

- Removed missing timestamps
- Dropped name/address/geographic text fields
- Removed irrelevant product attributes
- Dropped columns with no variance

Merging

- Joined all 6 tables into one analytical dataset
- Prepared features for segmentation

Preprocessing & Cleaning



Standardization

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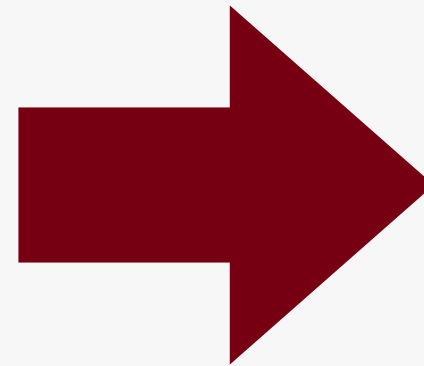
Our Dataset

Cluster-Then-Predict Framework:



Unsupervised Learning: Customer Segmentation

K-Means Clustering to
identify behavioral
customer personas.

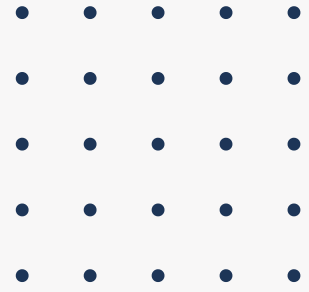


Supervised Learning: Predictive Models per Segment

Prediction Task 1:
Next-Purchase Category
(Classification)

Prediction Task 2:
Customer Churn
(Binary Classification)

Our Methods



1) K-Means Clustering

Feature Engineering (Customer-Level)

- Monetary: TotalSpend, AvgOrderValue, Spend per Category
- Frequency/Recency: NumTransactions, RecencyDays
- Basket Composition: AvgBasketSize, CategoryVariety
- Category Mix (% of spend across categories)

Preprocessing

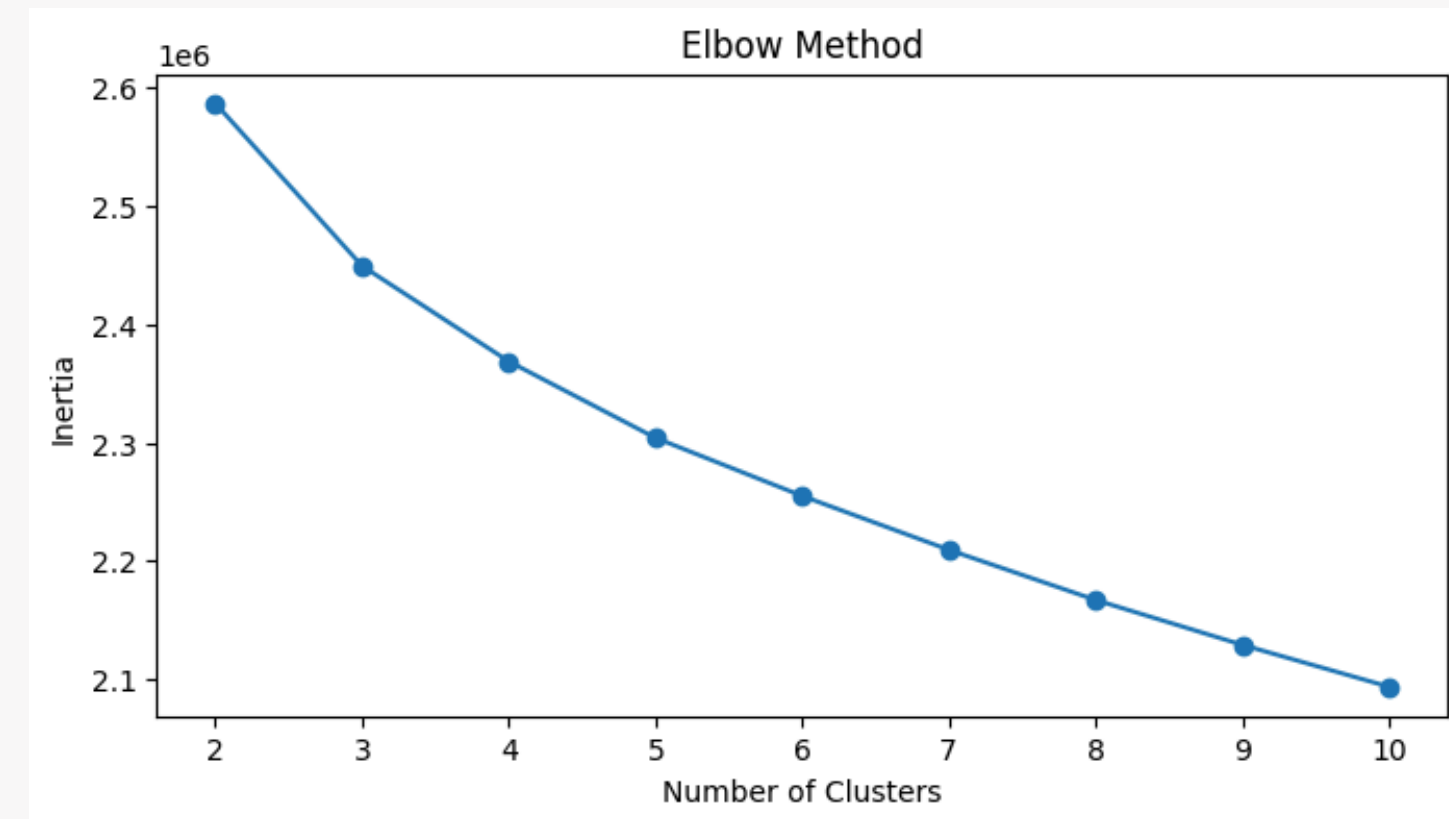
- Standardized all features (MinMax or StandardScaler)
- Removed outliers with extreme spend (>99.9%)

Choosing k

- Elbow method → kink around $k = 4$
- Silhouette score → peak at $k = 2$

Model

- Fit K-Means ($k = 4$)
- Attached cluster labels to each customer



Silhouette Score

$k=2$: 0.1506

**(highest, but too few clusters
not useful for segmentation)**

$k=3$: 0.1121

$k=4$: 0.0776

(acceptable & meaningful)

... higher k values drop heavily

Customer Personas

Cluster 0 — Steady General Shoppers

Medium spend (\$44,400) | Average Order (\$650) | Balanced basket (13 items)

→ steady frequent customers with balanced purchasing pattern

Cluster 1 — Focused High-Value Shoppers

High spend (\$53,600) | Average Order (\$848) | Higher basket (16 items)

→ lowest frequency, highest category concentration (Top1 = 0.206)

so slightly more category-focused

→ good for targeted category promos

Cluster 2 — Light / Price-Sensitive Shoppers

Lowest spend (\$15,900) | Average Order (\$240) | Smaller basket (4 items)

→ occasional visitors, buy essentials only with low spend across all categories

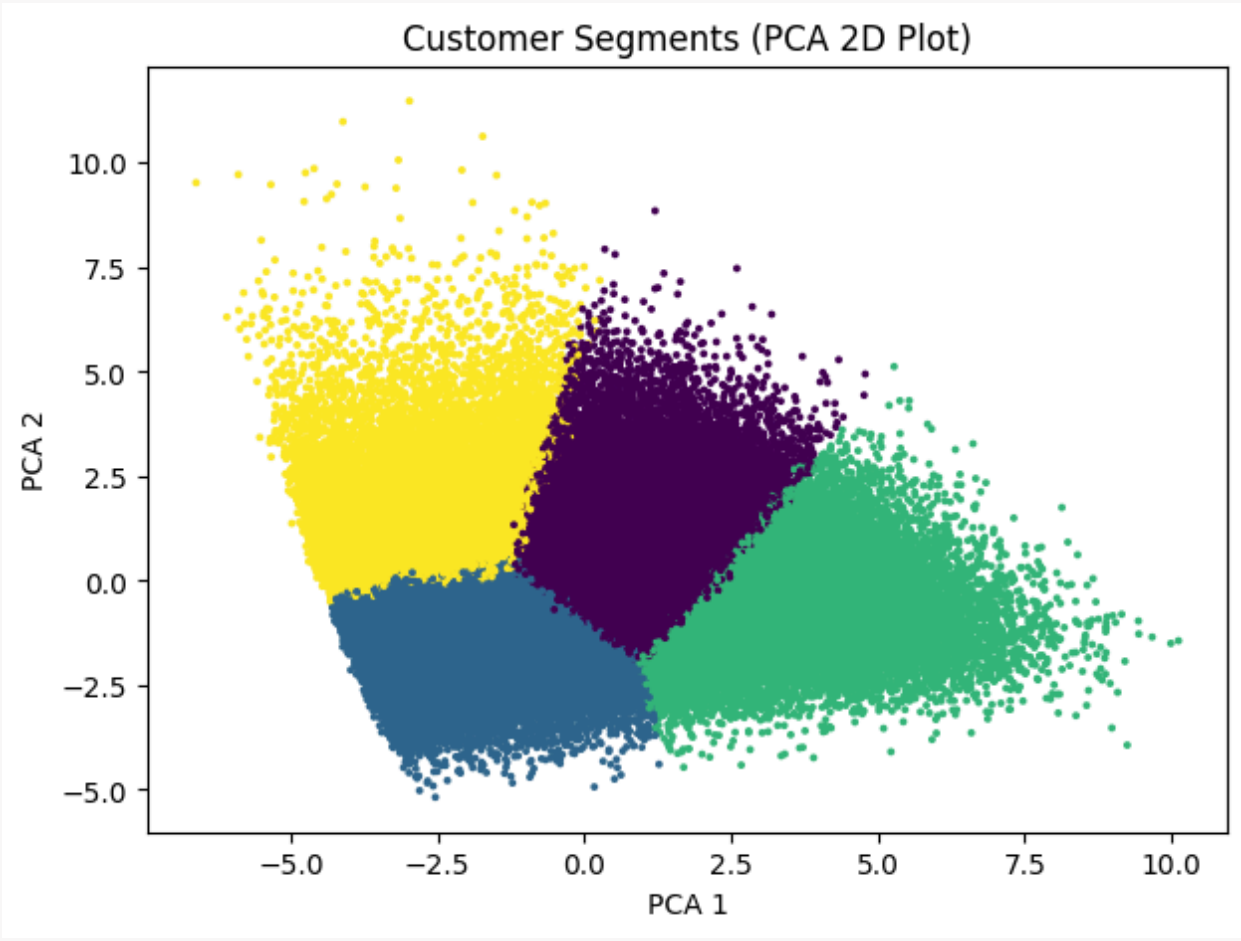
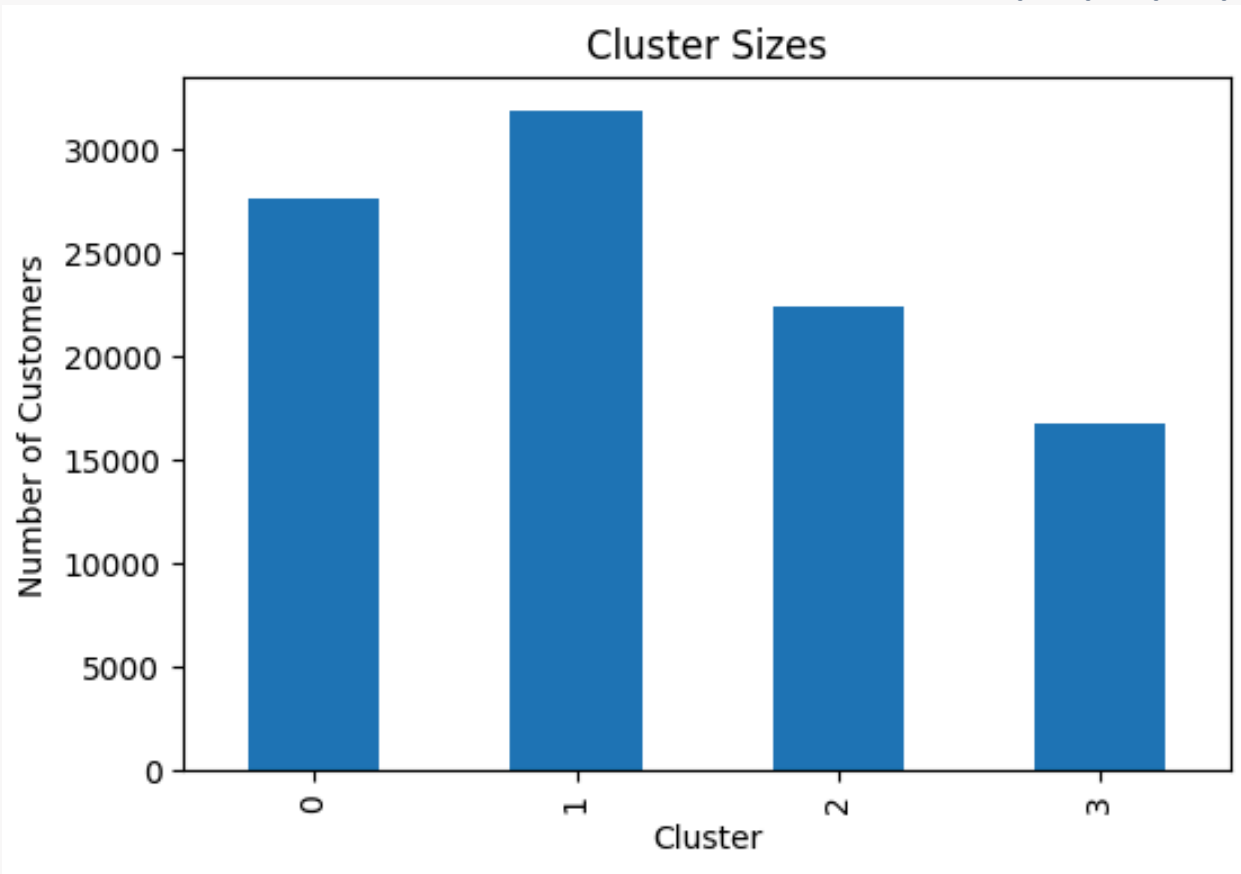
→ opportunity for basket-growth strategies

Cluster 3 — Premium Power Shoppers

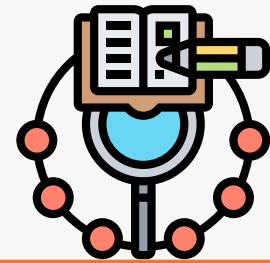
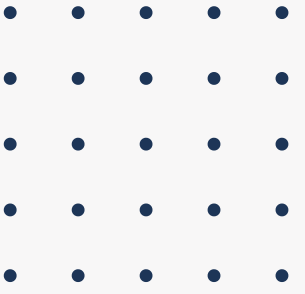
Highest spend (\$77,900) | Average Order (\$1102) | Largest baskets (21 items)

→ loyal, high-value customers across all categories (highest RFM)

	TotalSpend	AvgOrderValue	Frequency	RecencyDays	AvgTimeBetweenPurchases	AvgBasketSize	CategoryVariety
Cluster							
0	44422.040111	659.355833	68.188850	1.402755	1.278567	13.056430	10.983893
1	53618.419531	848.273647	63.763619	1.588821	1.521422	16.815873	10.886010
2	15899.688042	240.951898	66.779702	1.489501	1.352222	4.784358	10.956208
3	77979.103262	1102.716848	71.065814	1.353667	1.152219	21.473164	10.981675



2) Classification Model: Next-Purchase Category



Traditional Approach:

- **Train ONE model for ALL customers**
- **Ignores behavioral differences between segments**

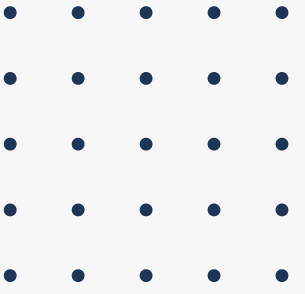


Our Approach:

- **Use customer segments from clustering result (4 clusters)**
- **Train separate models per segment**
- **Result: Higher accuracy through personalized modeling**

2) Classification Model: Next-Purchase Category

Feature Engineering



1. Purchase Sequence Features

Last 7 categories purchased → Captures buying patterns
Most important predictor of next purchase

2. Category Transition Probabilities

Probability matrix: "After buying X, what's likelihood of buying Y?"

Top transitions:

Beverages → Confections: 12.7%
Snails → Confections: 12.7%
Meat → Confections: 12.7%
Confections → Confections: 12.6% (repeat buyers)



3. Recency & Frequency

Days since last purchase
Average time between purchases
Category purchase frequency per customer

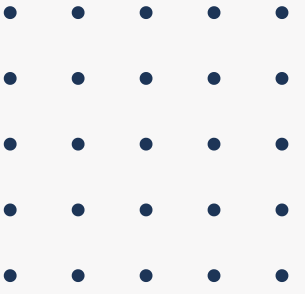
4. Spending Behavior

Current transaction spend
Average spend per customer
Spend ratio (current vs average)

5. Temporal Features

(Day of week, Month) Customer Characteristics

2) Classification Model: Next-Purchase Category



Model Tested

1. Logistic Regression (Baseline)

Fast, interpretable
Poor performance: ~12% accuracy
Cannot capture complex patterns

2. Random Forest

Ensemble of decision trees
Good performance: ~80–85% accuracy
Handles non-linear relationships

3. Extra Trees (Best Performer)

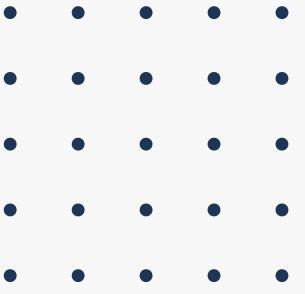
Similar to Random Forest but faster
Outstanding performance: 98–99% accuracy
Fully parallelized training

4. Naive Bayes

Probabilistic classifier
Fast but simplistic
Poor performance: ~10% accuracy

2) Classification Model: Next-Purchase Category

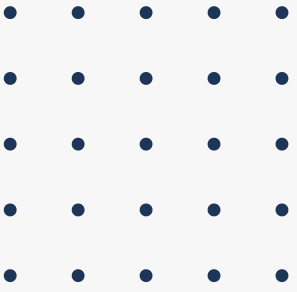
Global Models (All Customers Together)



Model	Accuracy	F1-Score
Logistic Regression	12.60%	2.80%
Random Forest	80.00%	77.80%
Extra Trees	98.90%	98.90%
Naive Bayes	11.00%	6.80%

 Best Global Model: Extra Trees (98.9%)

2) Classification Model: Next-Purchase Category



Results: Cluster-Specific Models

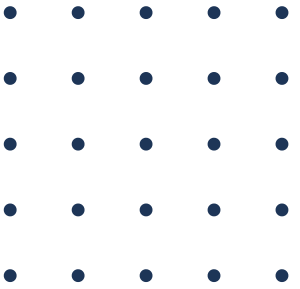
Cluster 0

Model	Accuracy	F1-Score
Logistic Regression	11.50%	2.40%
Random Forest	85.10%	83.90%
Extra Trees	98.60%	98.60%
Naive Bayes	10.40%	4.60%

Cluster 1

Model	Accuracy	F1-Score
Logistic Regression	15.50%	4.10%
Random Forest	83.50%	81.30%
Extra Trees	98.40%	98.30%
Naive Bayes	12.20%	6.90%

2) Classification Model: Next-Purchase Category



Results: Cluster-Specific Models

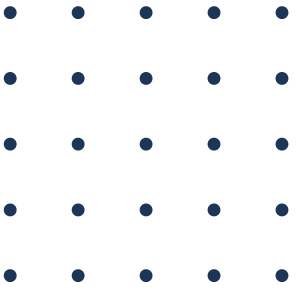
Cluster 2

Model	Accuracy	F1-Score
Logistic Regression	12.70%	2.90%
Random Forest	78.20%	75.70%
Extra Trees	99.30%	99.30%
Naive Bayes	10.80%	7.10%

Cluster 3

Model	Accuracy	F1-Score
Logistic Regression	12.10%	2.60%
Random Forest	82.10%	80.80%
Extra Trees	97.80%	97.80%
Naive Bayes	10.10%	6.10%

2) Classification Model: Next-Purchase Category



Key Findings: Feature Importance

Top 10 Most Predictive Features

1. TransitionProb (Category transition probability) ★★
2. Prev1 (Last category purchased)
3. Cat1Freq (How often customer buys that category)
4. SpendRatio (Current vs average spend)
5. AvgSpend (Customer's average spending)
6. CurrentSpend (This transaction's spend)
7. AvgDaysBetween (Purchase frequency)
8. TxnNumber (Position in purchase sequence)
9. Prev3 (3rd last category)
10. Prev2 (2nd last category)



Key Insight:

Recent purchase history +
transition patterns =
highly accurate
predictions

3) Predicting Customer Churn

Defined Churn

- Calculated Recency Days
- Tested cutoffs (7, 10, 14, 20, 30)
- Final threshold: Recency > 7 days

```
Cutoff 7 days , churn rate = 0.015
Cutoff 10 days , churn rate = 0.003
Cutoff 14 days , churn rate = 0.000
Cutoff 20 days , churn rate = 0.000
```

Build Global Churn Model

- Features: spending, frequency, basket size, category mix
- Models tested: Logistic Regression, Random Forest, Gradient Boosting
- Evaluated with 80/20 train-test split
- Random forest model hyperparameters are tuned with RandomizedSearchCV

Model Evaluation:

We prioritized:

- Recall, Precision, F1 score, ROC-AUC (the model's ability to separate churners vs non-churners across all threshold)

Not accuracy for churn since a model predicting that "everyone stays" could reach a high accuracy but detect no churners.

```
customer_features.groupby('Cluster')['Churn'].mean()
```

Churn	
Cluster	
0	0.012061
1	0.020162
2	0.016403
3	0.013430

3) Predicting Customer Churn

Logistic Regression

Accuracy : 0.6420
TPR/Recall: 0.7023
Precision : 0.0292
F1-score : 0.0561
ROC-AUC : 0.7487

Random Forest (Tuned)

Accuracy : 0.9849
TPR/Recall: 0.0000
Precision : 0.0000
F1-score : 0.0000
ROC-AUC : 0.7095

Gradient Boosting

Accuracy : 0.9841
TPR/Recall: 0.0134
Precision : 0.1739
F1-score : 0.0248
ROC-AUC : 0.7670

Logistic Regression:

- is able to catch some churners (**good recall**)
- but because churn is extremely rare, many of its churn predictions are false positives, giving it very **low precision**.
- Still, its **moderate ROC-AUC** shows it is actually learning meaningful ranking patterns.

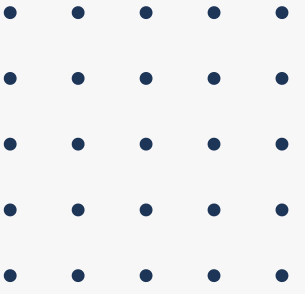
Boosting models are strong rankers and handle structured tabular data well. However, with this level of **class imbalance** and **limited churn signal**, its **precision and F1 remain low**.

Random Forest predicts no churners at all, so it never produces a true positive, which forces both **precision and recall to 0**.

This happens because:

- **Churn is extremely rare** ($\approx 1-3\%$), so the model defaults to predicting the majority class.
- Even with `class_weight="balanced"`, there still aren't enough churn examples to learn meaningful patterns.
- With **only January-May data**, churners look very similar to non-churners, giving the model **weak signal**.
- Predicting "everyone stays" yields 97-98% accuracy, which the model sees as optimal.

3) Predicting Customer Churn



We tried the Cluster-Then-Predict Method because we thought different customer segments might churn for different reasons, and that building segment-specific models could capture more personalized churn patterns.

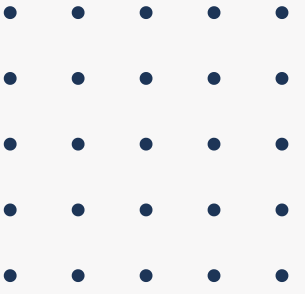
Steps we took:

- Using the original global train-test split, data was filtered by cluster.
- For each cluster, a separate Logistic Regression model and a Random Forrest were trained.
- Class weighting was applied to handle churn imbalance.

We stuck with the global model because:

- The dataset is highly imbalanced, and the global model has access to all churn cases, making learning more stable, **still not satisfactory**.
- **Splitting into clusters resulted in too few churn observations per segment, weakening model training.**

Results and Conclusions



1. Customer Segmentation

- Successfully identified 4 distinct customer personas
- Clear differentiation in spending, frequency, and basket behavior
- Actionable segments for targeted marketing

2. Next Purchase Prediction

- 99.3% accuracy in predicting next category
- Validated cluster-then-predict framework
- Identified key behavioral drivers (transition patterns)

3. Combined Insights

- Demonstrated value of segmentation for predictive modeling
- Cluster-specific models outperform global approaches
- Strong purchase sequence patterns across all segments

Academic Success vs Business Reality

As a Learning Project: ★★★★★

Demonstrated full ML workflow (EDA → Clustering → Classification)

Applied multiple algorithms and evaluated rigorously

Generated actionable business insights

Achieved high technical performance metrics

For Production Deployment: ⚠️ NOT READY

Critical Gap: Only 5 months of data (Jan-May 2018)

Why More Data is Essential

Current: 5 months = Limited snapshot

Needed: 12-24 months minimum for production

What's Missing:

Seasonal Patterns: Cannot detect holiday/annual trends

True Churn: 5 months insufficient to identify churned customers

Stability Testing: Model performance over time unknown